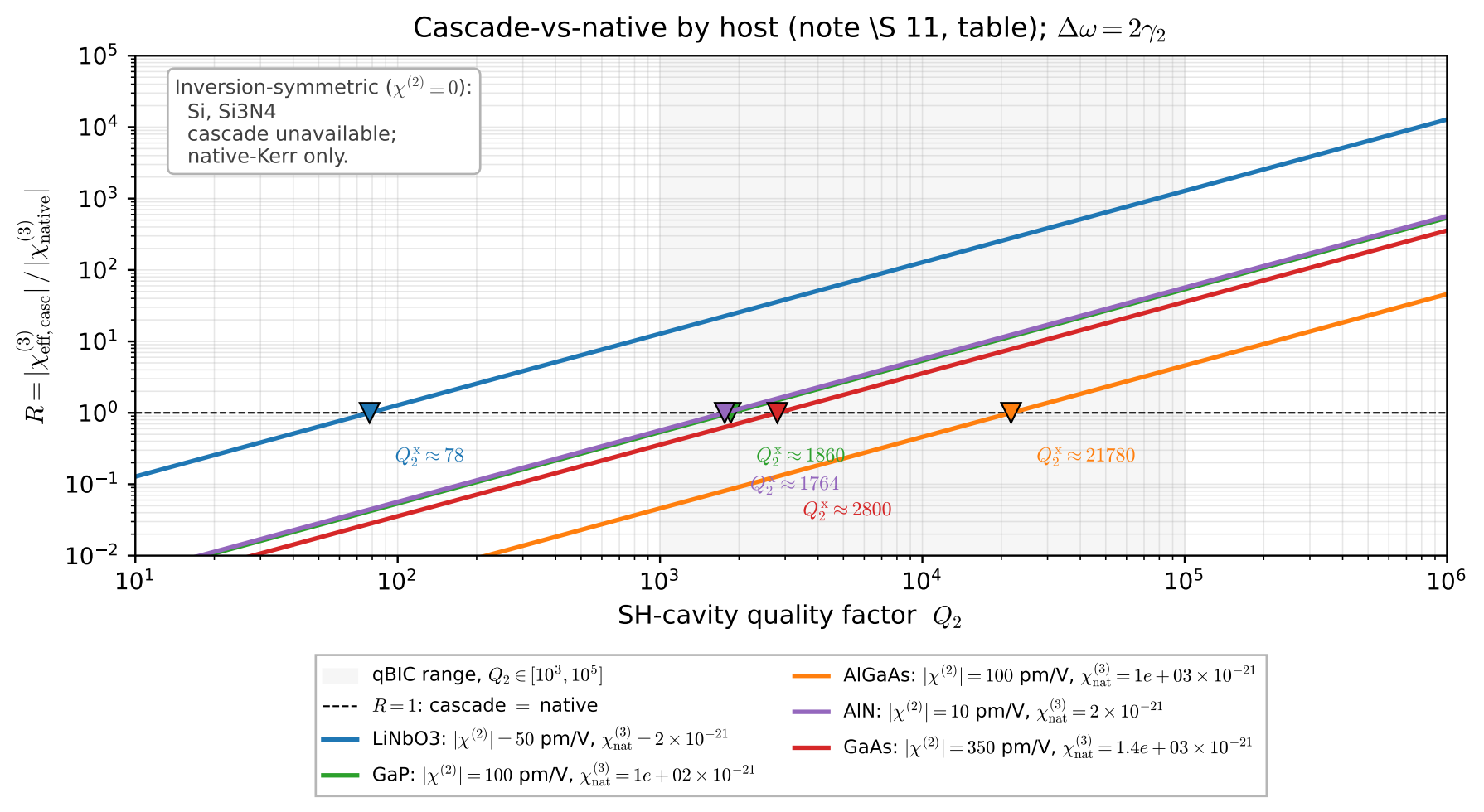
Cascade-vs-native by host (note \S 11, table); $\Delta\omega = 2\gamma_2$
Inversion-symmetric ($\chi^{(2)} \equiv 0$):
Si, Si3N4
cascade unavailable;
native-Kerr only.
$R = |\chi^{(3)}_{\mathrm{eff,\,casc}}| \,/\, |\chi^{(3)}_{\mathrm{native}}|$
SH-cavity quality factor Q_2
$Q_2^{\times} \approx 78$
$Q_2^{\times} \approx 1860$
$Q_2^{\times} \approx 1764$
$Q_2^{\times} \approx 2800$
$Q_2^{\times} \approx 21780$
qBIC range, $Q_2 \in [10^3, 10^5]$
$R = 1$: cascade = native
LiNbO3: $|\chi^{(2)}| = 50$ pm/V, $\chi^{(3)}_{\mathrm{nat}} = 2 \times 10^{-21}$
GaP: $|\chi^{(2)}| = 100$ pm/V, $\chi^{(3)}_{\mathrm{nat}} = 1e+02 \times 10^{-21}$
AlGaAs: $|\chi^{(2)}| = 100$ pm/V, $\chi^{(3)}_{\mathrm{nat}} = 1e+03 \times 10^{-21}$
AlN: $|\chi^{(2)}| = 10$ pm/V, $\chi^{(3)}_{\mathrm{nat}} = 2 \times 10^{-21}$
GaAs: $|\chi^{(2)}| = 350$ pm/V, $\chi^{(3)}_{\mathrm{nat}} = 1.4e+03 \times 10^{-21}$